

Soru-1 a)

$$h=0,1$$

$$f'(x_0) = \frac{1}{2h} \cdot [-25f(x_0) + 48f(x_0+h) - 36f(x_0+2h) + 16f(x_0+3h) - 3f(x_0+4h)]$$

$$f'(1,8) = 5 \cdot [-25f(1,8) + 48f(1,9) - 36f(2,0) + 16f(2,1) - 3f(2,2)]$$

$$f'(1,8) = 0,9853365 //$$

$$f'(x_0) = \frac{1}{2h} \cdot [f(x_0+h) - f(x_0-h)]$$

$$f'(1,9) = 5 \cdot [f(2,0) - f(1,8)] \quad f'(1,9) = 2,953165 //$$

$$f'(x_0) = \frac{1}{12h} \cdot [f(x_0-2h) - 8f(x_0-h) + 8f(x_0+h) - f(x_0+2h)]$$

$$f'(2,0) = \frac{5}{6} [f(1,8) - 8f(1,9) + 8f(2,1) - f(2,2)]$$

$$f'(2,0) = 2,249704083 //$$

$$f'(x_0) = \frac{1}{2h} [f(x_0+h) - f(x_0-h)]$$

$$f'(2,1) = 5 \cdot [f(2,2) - f(2,0)] \quad f'(2,1) = 1,8609585 //$$

$$f'(x_0) = \frac{1}{2h} [3f(x_0) - 4f(x_0-h) + f(x_0-2h)]$$

$$f'(2,2) = 5 \cdot [3 \cdot f(2,2) - 4f(2,1) + f(2,0)] \quad f(2,2) = 1,5188815 //$$

b) $h=0,1 \rightarrow f''(2) \cong \frac{1}{(0,1)^2} [f(1,9) - 2f(2,0) + f(2,1)] = 29,593200$

$$h=0,2 \rightarrow f''(2) \cong \frac{1}{(0,2)^2} [f(1,9) - 2f(2,0) + f(2,2)] = 29,704275$$

$$f''(x) = e^x(x+2) \quad f''(2) = e^2(2+2) = 29,556224$$

$$h=0,1 \text{ iken } \text{hata} = 3,6976 \cdot 10^{-2} // \quad h=0,2 \text{ iken } \text{hata} = 1,48051 \cdot 10^{-1} //$$

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Soru-2 $f(x) = e^x + 2^{-x} + 2\cos x - 6$, $f'(x) = e^x - 2^{-x} \cdot \ln 2 - 2\sin x$

$x_1 = 1$ alalım, $n=1$ için

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 1 - \frac{(-1,7011135538)}{0,68876626856} = 3,46973801052$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 3,46973801052 - \frac{24,3272653045}{32,7123789304} = 2,72612646919$$

$$|x_3 - x_2| = 0,74367154133 < 10^{-1} \text{ (devam)}$$

$$x_4 = x_3 - \frac{7,53488418047}{14,5711336638} = 2,20489834261 \text{ (devam)}$$

$$x_5 = x_4 - \frac{2,10132084412}{7,30777789284} = 1,917352421 \text{ (devam)}$$

$$x_6 = x_5 - \frac{0,98834246866}{4,7383243568} = 1,83539477962 \text{ (devam)}$$

$$x_7 = x_6 - \frac{0,02477937816}{4,14298267699} = 1,82941373164 \text{ (devam)}$$

$$x_8 = x_7 - \frac{0,0001235801}{4,10170699347} = 1,8293836027 //$$

$$|x_8 - x_7| = 0,0003012894 < 10^{-4} \text{ (sağlandı) //$$

Soru-3 → Dikdörtgen Metodu;

$$\int_{0,8}^{1,2} \frac{\sin 2x}{x^2} \cdot dx \quad h = \frac{b-a}{n} \quad h = 0,05 \quad n = 8 \quad 0,05 \sum_{i=0}^7 f(0,8 + 0,05i)$$

$$\cong 0,05 [f(0,8) + f(0,85) + f(0,9) + f(0,95) + f(1) + f(1,05) + f(1,1) + f(1,15)]$$

$$\cong 0,40547430362 //$$

→ Yamuklar Metodu;

$$\frac{0,05}{2} [f(0,8) + f(1,2)] + 2 \cdot [f(0,85) + f(0,9) + f(0,95) + f(1) + f(1,05) + f(1,1) + f(1,15)]$$

$$\cong (2,03092540791 + 13,0953046356) \cdot \frac{0,05}{2} \cong 0,37815525108 //$$

→ Simpson Metodu;

$$\cong \frac{0,05}{3} [f(0,8) + f(0,85) + f(0,9) + f(0,95) + f(1) + f(1,05) + f(1,1) + f(1,15) + f(1,2)]$$

$$\cong 22,6620000816$$

$$\int_{0,8}^{1,2} \frac{\sin 2x}{x^2} \cdot dx \cong 22,6620000816 \cdot \frac{0,05}{3} \cong 0,37770000136 //$$

Soru-4 Euler $\rightarrow h_1 = \frac{1}{40}, x_0 = 0, y_0 = 1, f(x, y) = xe^{-y} + \frac{x}{x^2 + 1}$

1. adım $\rightarrow x_1 = x_0 + h = 0 + \frac{1}{40} = \frac{1}{40}$

$y(x_1) = y(\frac{1}{40}) = y_1 = y_0 + h \cdot f(x_0, y_0) = 1 + h \cdot f(0, 1) = 1 + \frac{1}{40} \cdot (0) = 1$

2. adım $\rightarrow x_2 = x_1 + h = \frac{1}{40} + \frac{1}{40} = \frac{1}{20}$

$y(x_2) = y(\frac{1}{20}) = y_2 = y_1 + h \cdot f(x_1, y_1) = 1 + h \cdot f(\frac{1}{40}, 1) = 1 + \frac{1}{40} \cdot (0,034181370788)$

3. adım $\rightarrow x_3 = x_2 + h = \frac{1}{20} + \frac{1}{40} = \frac{3}{40}$ $= 1,000854534269$

$y(x_3) = y(\frac{3}{40}) = y_3 = y_2 + h \cdot f(x_2, y_2) = 1,000854534269 + h \cdot f(\frac{1}{20}, 1,000854534269)$
 $= 1,000854534269 + \frac{1}{40} \cdot (0,068253572213) = 1,002560873575$

4. adım $\rightarrow x_4 = x_3 + h = \frac{3}{40} + \frac{1}{40} = \frac{1}{10}$

$y(x_4) = y(\frac{1}{10}) = y_4 = y_3 + h \cdot f(x_3, y_3) = 1,002560873575 + h \cdot f(\frac{3}{40}, 1,002560873575)$
 $= 1,002560873575 + \frac{1}{40} \cdot (0,1021008763001) = 1,005113395482$

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38. adım $\rightarrow x_{38} = \frac{19}{20}$

$y_{38} = y_{37} + h \cdot f(x_{37}, y_{37}) = 1,42598598 + h \cdot f(\frac{37}{40}, 1,42598598)$
 $= 1,42598598 + \frac{1}{40} \cdot (0,72073542) = 1,440043$

39. adım $\rightarrow x_{39} = \frac{39}{40}$

$y_{39} = y_{38} + h \cdot f(x_{38}, y_{38}) = 1,4440043 + h \cdot f(\frac{19}{20}, 1,4440043)$
 $= 1,4440043 + \frac{1}{40} \cdot (0,72352183) = 1,46209248$

40. adım $\rightarrow x_{40} = 1$

$y_{40} = y_{39} + h \cdot f(x_{39}, y_{39}) = 1,46209248 + h \cdot f(\frac{39}{40}, 1,46209248)$
 $= 1,46209248 + \frac{1}{40} \cdot (0,7257968) = 1,48023740916 //$

Heun →

$$y(x_i+h) = y(x_i) + \left[\frac{k_1}{2} + \frac{k_2}{2} \right] \cdot h$$

$$k_1 = f(x_i, y_i)$$

$$k_2 = f(x_i+h, y_i+k_1 \cdot h)$$

$$u(0,05) = u(0) + \left[\frac{f(0,1)}{2} + \frac{f(0,05, 1+f(0,1) \cdot 0,05)}{2} \right] \cdot 0,05$$

$$u(0,05) = 1,00133539454861$$

$$u(0,55) = 1,20998252764249$$

$$u(0,1) = 1,00546679221003$$

$$u(0,6) = 1,25534405172682$$

$$u(0,15) = 1,01264477483615$$

$$u(0,65) = 1,30748990012339$$

$$u(0,2) = 1,02312036575215$$

$$u(0,7) = 1,36487806622654$$

$$u(0,25) = 1,03714507578677$$

$$u(0,75) = 1,42836749983928$$

$$u(0,3) = 1,05497094942042$$

$$u(0,8) = 1,4982181551882$$

$$u(0,35) = 1,07685061106402$$

$$u(0,85) = 1,57469103925913$$

$$u(0,4) = 1,103037311148118$$

$$u(0,9) = 1,6580482603272$$

$$u(0,45) = 1,13378497436659$$

$$u(0,95) = 1,74855307683463$$

$$u(0,5) = 1,16934824309339$$

$$u(1) = u(0,95) + \left[\frac{k_1}{2} + \frac{k_2}{2} \right] \cdot 0,05 = 1,84646934654625 //$$

Runge-Kutta → $y(x_i+h) = y(x_i) + \frac{1}{6}(k_1+2k_2+2k_3+k_4) \cdot h$ $h=0,1$

$$k_1 = f(x_i, y_i) \quad k_2 = f(x_i + \frac{1}{2} \cdot h, y_i + \frac{1}{2} \cdot k_1 \cdot h)$$

$$k_3 = f(x_i + \frac{1}{2} \cdot h, y_i + \frac{1}{2} \cdot k_2 \cdot h) \quad k_4 = f(x_i + h, y_i + k_3 \cdot h)$$

$$u(0,1) = u(0) + \frac{1}{6} \cdot (k_1 + 2k_2 + 2k_3 + k_4) \cdot 0,1$$

$$k_1 = f(0,1) \quad , \quad k_2 = f(0 + \frac{1}{2} \cdot 0,1, 1 + \frac{1}{2} \cdot k_1 \cdot 0,1)$$

$$k_3 = f(0 + \frac{1}{2} \cdot 0,1, 1 + \frac{1}{2} \cdot k_2 \cdot 0,1) \quad k_4 = f(0 + 0,1, 1 + k_3 \cdot 0,1) \Rightarrow$$

$$u(0,1) = 1,00998$$

$$u(0,2) = u(0,1) + \frac{1}{6} \cdot (k_1 + 2k_2 + 2k_3 + k_4) \cdot 0,1 = 1,03961$$

$$u(0,3) = 1,08803$$

$$u(0,4) = 1,15421$$

$$u(0,5) = 1,23657$$

$$u(0,6) = 1,33374$$

$$u(1) = 1,84657 //$$

$$u(0,7) = 1,44439$$

$$u(0,8) = 1,56735$$

$$u(0,9) = 1,70166$$

Soru-5

$$4x_1 - x_2 + 2x_3 = 0 \rightarrow x_1 = \frac{x_2 - 2x_3}{4}$$

$$-x_1 + 6x_2 - x_3 = 1 \rightarrow x_2 = \frac{1 + x_1 + x_3}{6}$$

$$2x_1 - x_2 + 5x_3 = -8 \rightarrow x_3 = \frac{-8 - 2x_1 + x_2}{5}$$

$$\text{Bastlangıç} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

1. iterasyon:

$$x_1 = \frac{0 - 2(0)}{4} = 0$$

$$x_2 = \frac{1 + 0 + 0}{6} = 0,1666$$

$$x_3 = \frac{-8 - 2(0) + (0,1666)}{5} = -1,5666$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0,1666 \\ -1,5666 \end{pmatrix}$$

$$|x_{3\text{son}} - x_{3\text{önce}}| > 10^{-3} \text{ (devam)}$$

2. iterasyon

$$x_1 = 0,8249$$

$$x_2 = 0,0430$$

$$x_3 = -9,6068$$

$$|x_{3\text{son}} - x_{3\text{önce}}| > 10^{-3} \text{ (devam)}$$

3. iterasyon

$$x_1 = 4,9109$$

$$x_2 = -0,6159$$

$$x_3 = -3,6875 \text{ (devam)}$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 4,9109 \\ -0,6159 \\ -3,6875 \end{pmatrix}$$

4. iterasyon

$$x_1 = 1,3867$$

$$x_2 = -0,1663$$

$$x_3 = -2,3091$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1,3867 \\ -0,1663 \\ -2,3091 \end{pmatrix} \text{ (devam)}$$

5. iterasyon

$$x_1 = 1,1129$$

$$x_2 = -0,0327$$

$$x_3 = -2,0512$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1,1129 \\ -0,0327 \\ -2,0512 \end{pmatrix} \text{ (devam)}$$

6. iterasyon

$$x_1 = 1,0176$$

$$x_2 = -0,0056$$

$$x_3 = -2,0081$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1,0176 \\ -0,0056 \\ -2,0081 \end{pmatrix} \text{ (devam)}$$

7. iterasyon

$$x_1 = 1,0026$$

$$x_2 = -0,0009$$

$$x_3 = -2,0012$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1,0026 \\ -0,0009 \\ -2,0012 \end{pmatrix}$$

8. iterasyon

$$x_1 = \frac{(-0,0009) - 2 \cdot (-2,0012)}{4} = 1,0003$$

$$x_2 = \frac{1 + (1,0003) + (-2,0012)}{6} = -0,0001$$

$$x_3 = \frac{-8 - 2(1,0003) + (-0,0001)}{5} = -2,0001$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1,0003 \\ -0,0001 \\ -2,0001 \end{pmatrix} //$$

$$|x_{3\text{son}} - x_{3\text{önce}}| \approx 10^{-3}$$